Documentation for Surface Table Structure

This documentation outlines the current structure of the surface table, the purpose of its columns, and plans for future enhancements. The goal is to create a scalable and efficient system that supports finite element analysis (FEA) and can be extended as the project evolves.

Current Surface Table Structure

The surface table currently includes the following columns:

Column	Description	Purpose
local_id	Unique identifier for each surface.	Ensures every surface is uniquely identifiable.
node_1	Node ID for the first corner of the surface.	Defines the surface geometry.
node_2	Node ID for the second corner of the surface.	Defines the surface geometry.
node_3	Node ID for the third corner of the surface.	Defines the surface geometry.
node_4	Node ID for the fourth corner of the surface (or NaN for triangles).	Defines the surface geometry.
stringer_1	Stringer index corresponding to node_1.	Tracks surface alignment with structural elements.
stringer_2	Stringer index corresponding to node_2.	Tracks surface alignment with structural elements.
rib_1	Rib index corresponding to node_1.	Tracks surface alignment with structural elements.
rib_2	Rib index corresponding to node_2.	Tracks surface alignment with structural elements.
tags	Tag describing the surface type (quad, tri, irregular).	Facilitates categorization and downstream processing.

Column	Description	Purpose
area	Precomputed area of the surface.	Supports validation and mesh quality analysis.
aspect_ratio	Aspect ratio of the surface (max/min diagonal for quads or edges for triangles).	Ensures surface quality.

Current Implementation Notes

1. **Simplified Structure:**
   - Focuses on essential columns for geometry definition and quality analysis (area and aspect ratio).
   - Tags help categorize surfaces (e.g., quad, tri) for specialized handling.
2. **Quality Validation:**
   - Surface area and aspect ratio are included to detect and avoid degenerate or poorly shaped surfaces.

Planned Enhancements

In the future, the surface table can be extended to include the following columns to better support FEA workflows and integration with tools like Patran/Nastran:

1. Connectivity to Adjacent Surfaces

- Columns: adjacent_surface_1, adjacent_surface_2, ...
- Purpose: Tracks adjacent surfaces for connectivity validation and boundary continuity.
- Example: adjacent_surface_1 stores the local_id of a neighboring surface that shares an edge.

2. Material Properties

- Column: material_id.
- Purpose: Links surfaces to material properties for analysis (e.g., stiffness, density).

3. Element Type

- **Column: element_type.**
- **Purpose: Specifies the finite element type (e.g., CQUAD4, CTRIA3).**

4. Boundary Conditions

- **Columns: bc_edge_1, bc_edge_2, ...**
- **Purpose: Tracks fixed or constrained edges for boundary condition application.**

5. Thickness

- **Column: thickness.**
- **Purpose: Defines shell element thickness for structural analysis.**

6. Export Compatibility

- **Columns specific to Patran/Nastran format for easier export and simulation setup.**
- **Example: patran_element_id.**

Rationale for Enhancements

1. **Connectivity to Adjacent Surfaces:**
    - **Ensures continuity in the mesh and prevents gaps or overlaps in the finite element model.**
2. **Material Properties and Thickness:**
    - **Enables simulation-ready surfaces with all physical properties defined.**
3. **Boundary Conditions:**
    - **Facilitates direct mapping of constraints during FEA setup.**
4. **Export Compatibility:**
    - **Reduces preprocessing effort when transitioning to Patran/Nastran or similar tools.**

Steps for Future Implementation

1. **Current Implementation:**

   - **Ensure accurate calculation of area and aspect_ratio.**
   - **Validate surfaces during creation using these metrics.**
2. **Future Implementation:**
   - **Develop a connectivity algorithm to populate adjacent surface columns.**
   - **Extend table structure to include material properties, element type, and other attributes.**
   - **Integrate Patran/Nastran export functionality.**